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INTERNSHIP REPORT CYBER SECURITY (SOC)

Week 4 · Task 11 — Ingest Logs into SIEM + Basic Searches

1. Introduction

Security Information and Event Management (SIEM) is a security solution used to collect, monitor, and analyze security logs from different systems and devices. It helps security analysts identify suspicious activities and investigate potential security incidents.

2. Objective

The objective of this task is to understand SIEM log ingestion, learn basic search queries, identify failed login events, and understand how security events can be filtered and analyzed.

3. Tools and Technologies

- Splunk
- Wazuh
- Windows and Linux Logs
- TryHackMe – Introduction to SIEM

4. Log Ingestion

Log ingestion is the process of collecting and importing log data into a SIEM platform.

Common log sources include:

- Windows Security Logs
- Linux Authentication Logs
- System Events
- Application Logs

5. Basic Search Queries

Search queries are used to find specific information from collected log data.

Query 1: Search All Events

index=main

This query searches for events stored in the main index.

Query 2: Search Failed Events

index=main | search failed

This query filters events containing the word "failed".

6. Failed Login Analysis

Failed login events occur when an authentication attempt is unsuccessful. These events can help identify repeated login failures and possible unauthorized access attempts.

Example search:

index=main "failed"

To count matching events:

index=main "failed" | stats count

The stats command is used to calculate the number of matching events.

7. Time-Based Filtering

Time-based filtering allows analysts to examine security events within a selected period.

Common time ranges include:

- Last 15 minutes
- Last 1 hour
- Last 24 hours

8. Key Learnings

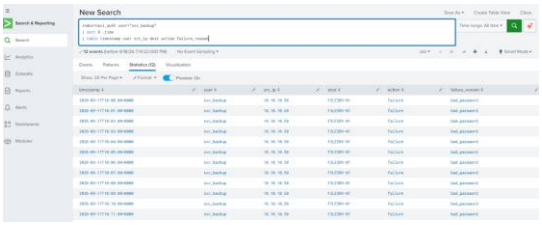
- Understanding the purpose of SIEM.
- Learning the concept of log ingestion.
- Understanding Windows and Linux log sources.
- Learning basic Splunk SPL search syntax.
- Understanding failed login event analysis.
- Learning how to count matching events.
- Understanding time-based event filtering.

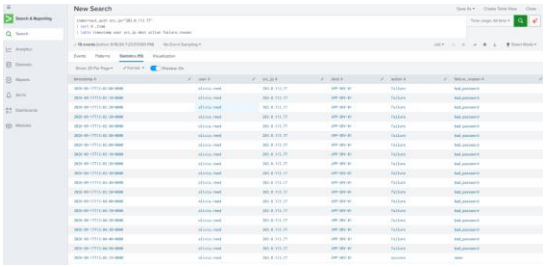
9. Conclusion

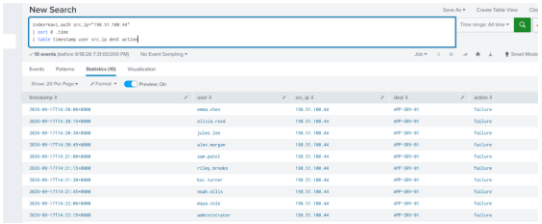
SIEM plays an important role in cybersecurity by providing centralized log monitoring and security event analysis. Understanding log ingestion, search queries, failed login filtering, and time-based analysis helps build a foundation for security monitoring and incident investigation.

10. Screen Shoots





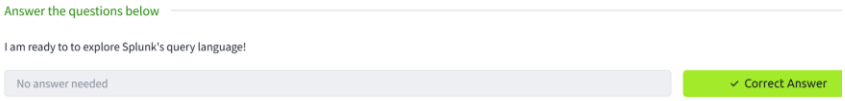




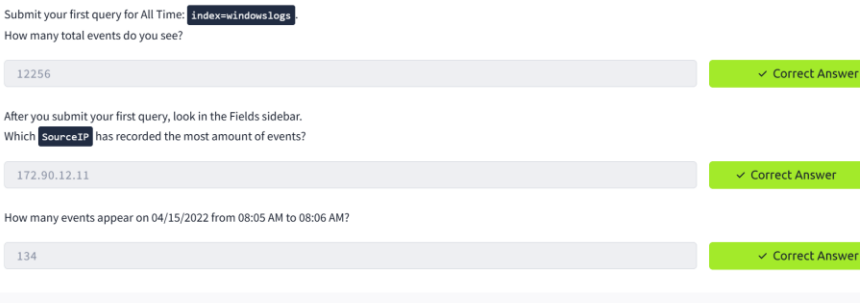
TRY HACK ME



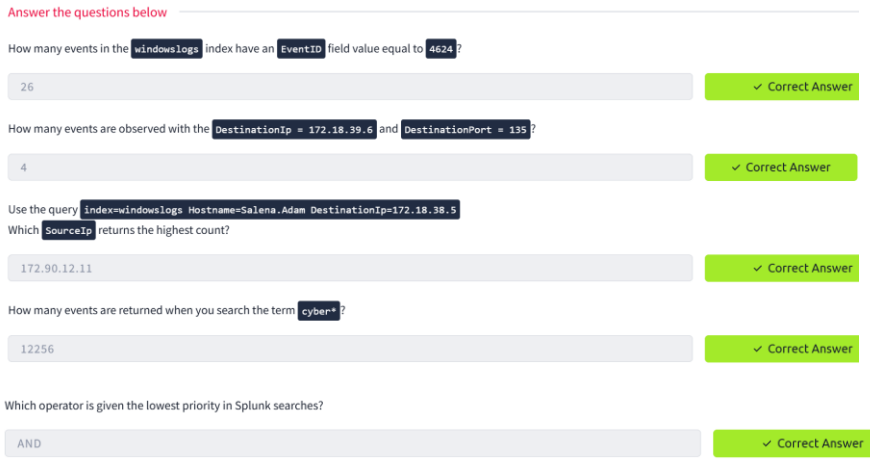
T1


Answer the questions below
I am ready to to explore Splunk's query language!
No answer needed
✓ Correct Answer

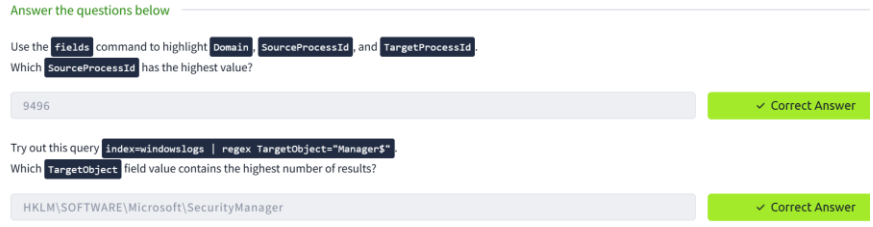
T2


Submit your first query for All Time: index=windowslogs.
How many total events do you see?
12256
✓ Correct Answer
After you submit your first query, look in the Fields sidebar.
Which SourceIP has recorded the most amount of events?
172.90.12.11
✓ Correct Answer
How many events appear on 04/15/2022 from 08:05 AM to 08:06 AM?
134
✓ Correct Answer

T3


Answer the questions below
How many events in the windowslogs index have an EventID field value equal to 4624?
26
✓ Correct Answer
How many events are observed with the DestinationIp = 172.18.39.6 and DestinationPort = 135?
4
✓ Correct Answer
Use the query index=windowslogs Hostname=Salena.Adam DestinationIp=172.18.38.5
Which SourceIp returns the highest count?
172.90.12.11
✓ Correct Answer
How many events are returned when you search the term cyber*?
12256
✓ Correct Answer
Which operator is given the lowest priority in Splunk searches?
AND
✓ Correct Answer

T4


Answer the questions below
Use the fields command to highlight Domain, SourceProcessId, and TargetProcessId.
Which SourceProcessId has the highest value?
9496
✓ Correct Answer
Try out this query index=windowslogs | regex TargetObject="Manager$".
Which TargetObject field value contains the highest number of results?
HKLM\SOFTWARE\Microsoft\SecurityManager
✓ Correct Answer

T5

Answer the questions below

Build a table that highlights the `EventID`, `AccountName`, and `AccountType` fields.
Which `AccountName` appears first in your results?

SYSTEM ✓ Correct Answer

Append the above query to include the `reverse` command.
Which `EventID` appears first?

800 ✓ Correct Answer

Use the query below to build a timeline of events.
What password was given to the user `Alberto`?

```
index=windowslogs EventID=1
| table _time ParentProcessId ProcessId ParentCommandLine CommandLine
| reverse
```

paw0rd1 ✓ Correct Answer

T6

Answer the questions below

Use the `top` command to query the `Image` field.
Which `Image` field value has the most occurrences?

C:\windows\system32\svchost.exe ✓ Correct Answer

Try out the `iplocation` command with the `SourceIp` field.
Which `Region` do the IP addresses in your events originate from?

California ✓ Correct Answer

Try out this `lookup` query. Which `Image` field value has the highest `RiskScore`?

```
index=windowslogs
| lookup image_riskscore Image OUTPUT RiskScore
| stats count by Image RiskScore
| sort - RiskScore
```

C:\Windows\System32\WindowsPowerShell\v1.0\powershell.exe ✓ Correct Answer

T7

Answer the questions below

Run the first anomaly detection query (**index=vpnlogs**).
Which other user is marked as an outlier?

jsmith ✓ Correct Answer

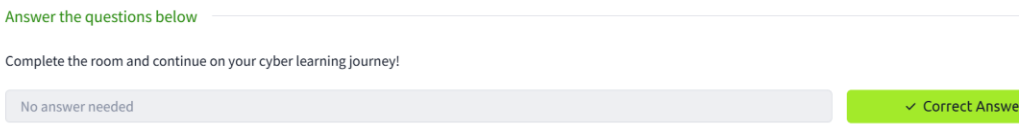
Which country is anomalous for the user from Q1?
Answer Example: CN

JP ✓ Correct Answer

Run the second anomaly detection query (**index=vpnlogs**).
Which user suspiciously logged in at 3 AM?

njackson ✓ Correct Answer

T8



Room completed: 100%

Task 1: Introduction

Splunk is a Security Information and Event Management (SIEM) platform used to collect, search, monitor, and analyze machine-generated data. It helps security analysts investigate logs, identify suspicious activities, and detect potential security threats.

Search Processing Language (SPL) is used in Splunk to search and analyze log data through commands, functions, and operators.

Task 2: Search & Reporting

The Search & Reporting App is the default interface used to search and analyze data in Splunk.

Its main features include:

- **Search Head:** Allows analysts to enter search queries.
- **Time Picker:** Selects the time range for searching events.
- **Search History:** Displays previously used search queries.
- **Data Summary:** Provides information about available hosts, sources, and sourcetypes.
- **Fields Sidebar:** Displays selected and interesting fields from log events.

The index=windowslogs query is used to search Windows log events. Analysts can use the fields sidebar to examine values such as usernames and source IP addresses.

Task 3: Search Operators

Search operators help filter and refine log data according to specific conditions.

Important operators include:

- **Equals (=):** Finds events where a field matches a specified value.
- **Not Equal (!=):** Excludes events with a particular value.
- **Greater Than (>):** Finds values greater than a specified number.
- **Less Than (<):** Finds values below a specified number.

- **AND:** Combines conditions that must both be true.
- **OR:** Finds events matching either condition.
- **NOT:** Excludes events matching a condition.
- **IN:** Searches for multiple specified values.
- **Wildcard (*):** Matches partial values or text patterns.

Quotation marks are used for exact phrases, while parentheses group conditions and control search logic.

Task 4: Filtering Results

Splunk provides filtering commands that help analysts organize and narrow down search results.

Important commands include:

- **Fields:** Includes or excludes selected fields from results.
- **Dedup:** Removes duplicate values.
- **Rename:** Changes field names to make results easier to understand.
- **Regex:** Filters events using regular expression patterns.

These commands help analysts focus on relevant information and make log analysis more efficient.

Task 5: Structuring Results

Structuring commands organize search results into a more readable format.

- **Table:** Displays selected fields in a tabular format.
- **Reverse:** Reverses the order of search results.
- **Sort:** Arranges results in ascending or descending order.

These commands help analysts review events, identify relevant information, and understand the sequence of activities.

Task 6: Transforming Commands

Transforming commands convert raw log events into useful summaries and statistics.

Important commands include:

- **Top:** Displays the most frequently occurring field values.
- **Rare:** Displays the least frequently occurring values.

- **Stats:** Calculates statistics such as count, average, maximum, minimum, and sum.
- **Chart:** Creates statistical tables that can be used for visualizations.
- **Timechart:** Displays event patterns over time.
- **Iplocation:** Adds geographical information about IP addresses.
- **Lookup:** Enriches events using external lookup tables.
- **Eval:** Creates or modifies fields and performs calculations.
- **Highlight:** Highlights selected values in raw event data.

These commands help security analysts identify patterns, summarize large datasets, and investigate unusual activities.

Task 7: Anomaly Detection

Anomaly detection is the process of identifying unusual activities that differ from normal behavior. In cybersecurity, it can help identify suspicious login attempts and potentially compromised accounts.

Splunk uses commands such as eventstats, eval, and where to analyze login patterns and identify outliers.

Two important anomaly detection methods covered in this task are:

1. Detecting Outliers by Country

This method compares a user's login activity across different countries. A login from an unusual country may indicate suspicious activity and requires further investigation.

2. Detecting Outliers by Hour

This method analyzes login times to identify activities occurring outside a user's normal login hours.

The concept of impossible travel was also introduced. It identifies potentially suspicious logins from geographically distant locations within an unusually short period.

Task 8: Conclusion

The Splunk: Exploring SPL room introduced important concepts of searching, filtering, structuring, and analyzing log data using Search Processing Language. It covered search operators, filtering commands, transforming commands, and anomaly detection techniques.

This learning helped build an understanding of how SIEM analysts use SPL to investigate security events, identify unusual login behavior, analyze IP addresses, and recognize potential

security threats. These concepts are useful for security monitoring, log investigation, and Security Operations Center (SOC) activities.